

Safe inference and the stability-controlled quasi-experiment (SCQE)

PS200C, Spring 2026

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Roadmap

- 1 Motivation: “safe inference” as a posture
- 2 The SCQE idea and identification
- 3 Applications: dexamethasone, HCQ, remdesivir
- 4 Relationship to other strategies (DID, SOO+ sensitivity)
- 5 Wrap-up

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Reaction 2: “Make a heroic assumption, call the estimate suggestive.”

- Covariate adjustment + no-unobserved-confounders is the classic case, but the same goes for IV, DID, synth, etc.
- Even calling the results “suggestive” is dangerous... helpful treatments may look harmful, vice versa.

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Today: a different identification strategy, designed to be argued about transparently.

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This baseline trend δ is the only assumption required to get the ATT.

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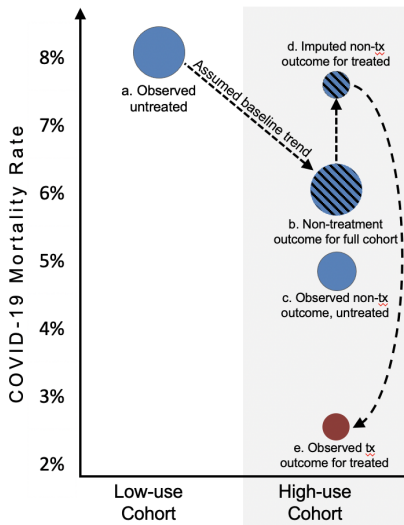
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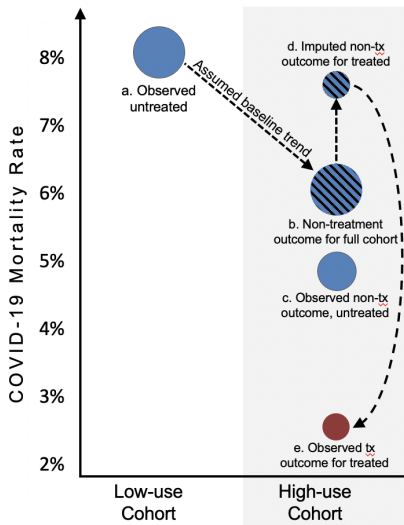
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Don't worry we won't need point values for δ ...there will be good uses for (i) defensible ranges, (ii) unbelievable ranges, or even just (iii) ranges where you cannot be sure either way...all will contribute to characterizing what we can(not) conclude.

How does δ +data tell you the ATT?

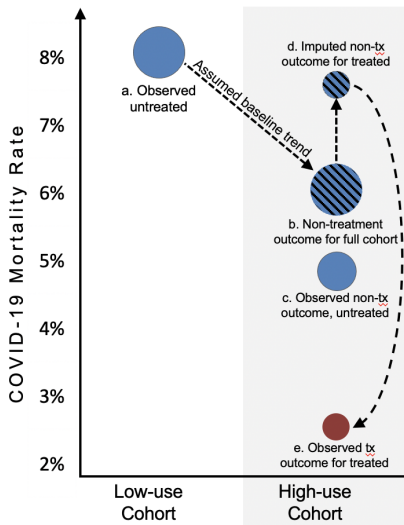


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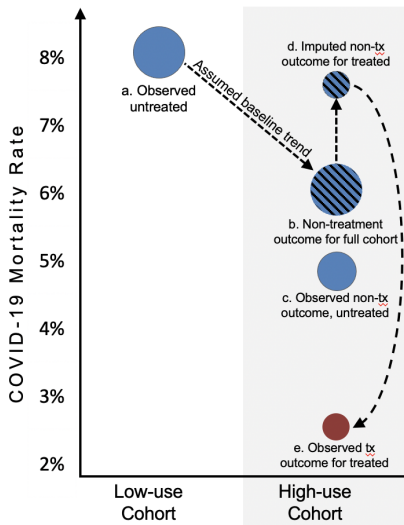
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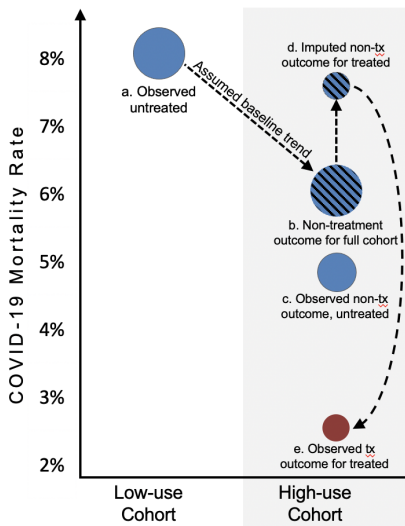
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- Then you can back out $\mathbb{E}[Y(0)|T = 1, D = 1]$ (d).
- We observe $\mathbb{E}[Y(1)|T = 1, D = 1]$ (e), and that minus (d) is the ATT.

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So if you want the ATT at $T = 1$,

$$\begin{aligned} ATT &\equiv \mathbb{E}[Y(1)|D = 1, T = 1] - \mathbb{E}[Y(0)|D = 1, T = 1] \\ &= \mathbb{E}[Y|D = 1, T = 1] - \left(\frac{\mathbb{E}[Y|T = 0] - \mathbb{E}[Y|D = 0, T = 1](1 - \pi_1) + \delta}{\pi_1} \right). \end{aligned} \quad (4)$$

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- The two “cohorts” could be different facilities, eras, or sites — whatever defines a sharp shift in treatment probability, as long as you can reason about non-treatment outcome differences between them.
- In fact you could do this with just one study cohort, and any guess of $\mathbb{E}[Y(0) \mid \text{your study sample}]$ (more soon)

Connection to IV: time as the instrument

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Concretely: define a pseudo-outcome $\tilde{Y}_i \equiv Y_i - \delta T_i$. Then under SCQE assumptions $\tilde{Y}_i(0) \perp\!\!\!\perp T_i$, and the Wald estimator gives:

$$ATT = \frac{\mathbb{E}[Y|T=1] - \mathbb{E}[Y|T=0] - \delta}{\pi_1}.$$

...which is just a (simpler looking) rearrangement of the SCQE estimator in (4).

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Worked example: dexamethasone for COVID-19

Cohorts:¹

- “Low-use” (first) cohort: days 44–101, **5.7%** took dexamethasone
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On its surface: big uptick in treatment, big drop in mortality...promising but could it just be the baseline trend?

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Before looking at the SCQE results, let's think about where δ might be.

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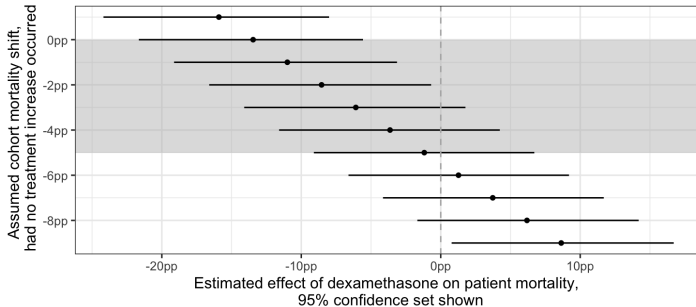
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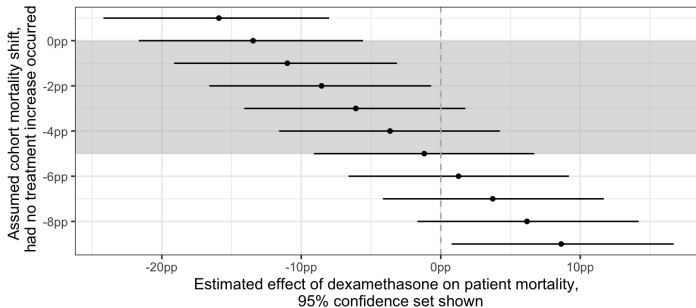
“All lines of evidence suggest a possible drop in baseline mortality. Taking these considerations collectively, and maintaining a conservative attitude, we postulated that a plausible range for the baseline trend ranged from no change to a reduction of up to 5pp (–46%).”

Dexamethasone: results



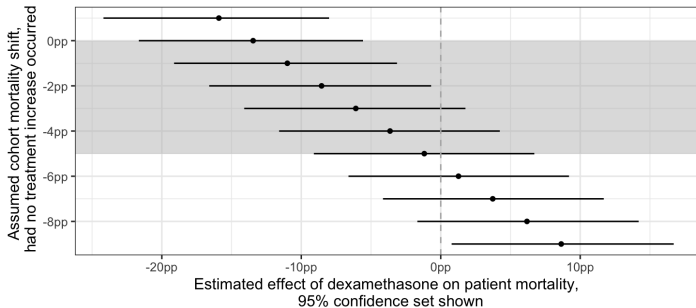
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- Beneficial ($p < 0.05$) if baseline mortality increased, flat, or fell by up to 2.3pp (−21%)
- Non-harmful point estimate until a drop of 5pp (−46%)
- Harmful ($p < 0.05$) only if baseline mortality fell by 6.8pp (−93%), leaving mortality of just 0.4%

Dexamethasone: what we concluded

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Consistent with the eventual results of the RECOVERY-trial.

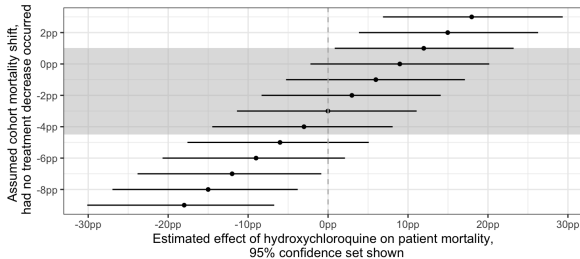
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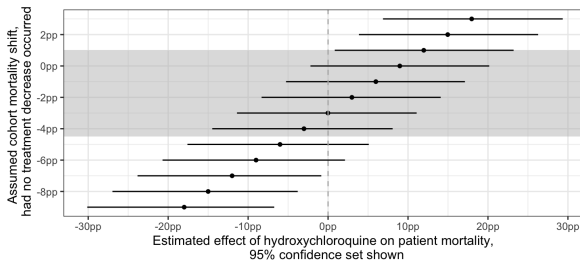
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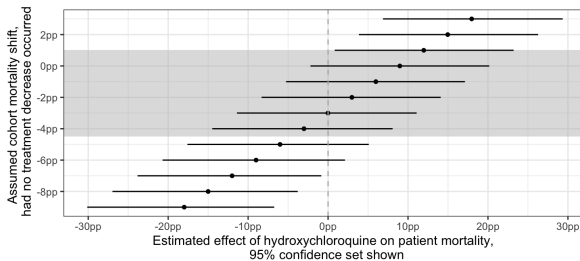


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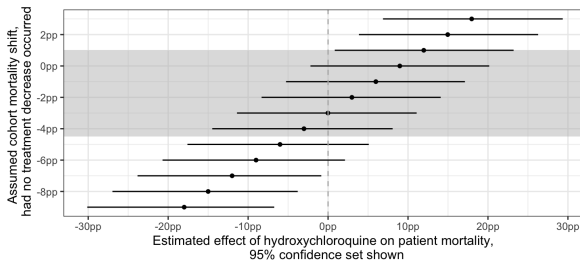
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Sig. beneficial only if baseline improved by $\geq 6.7\text{pp}$ (−46%); not credible.

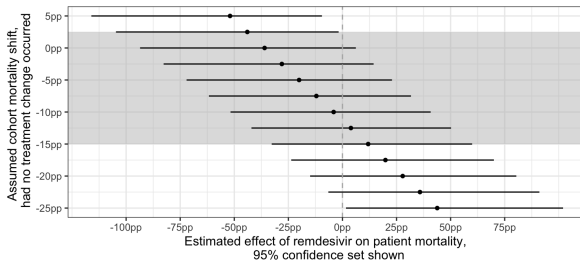
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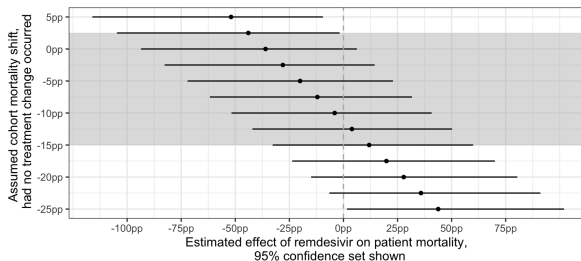
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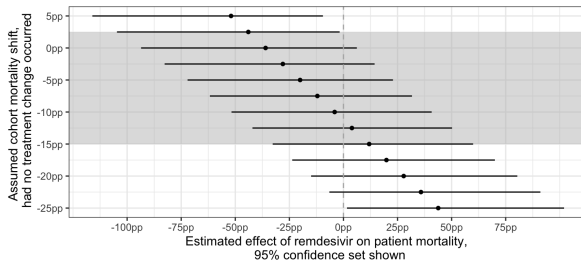


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Can rule out statistically significant harm – would imply 0% counterfactual mortality.

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Not precise, but a long way from either “we can’t say anything without an RCT” or “the observational study is our best guess.”

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- Impossible in cases like this – you cannot label patients in the first cohort as “would-be treated”.

Case 2: You can define those groups.

- Parallel trends = the two groups share a trend in $\mathbb{E}[Y(0)]$.
- Suppose you just look at change in mean outcomes in the control group, and you assume that is exactly δ .
- That is DID! So **DID is a special case of SCQE**, with a particular assumption about δ .
- SCQE shows DID as one estimate in a range, indexed by δ .

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Different assumptions offer different leverage... use what works.

Outline

- 1 Motivation: safe inference
- 2 The SCQE idea and identification
- 3 Applications
- 4 Relationship to other strategies
- 5 **Wrap-up**

Software

R package `scqe`: <https://github.com/chadhazlett/scqe>

- Two-cohort and one-cohort modes (full data or summary stats)
- ATT estimates and CIs over a range of δ
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You can plug in just cohort-by-treated numbers; doesn't require individual data..

Where to use SCQE

Medicine:

- While we wait for RCTs — present observational results in ways that prevent over-confidence
- Therapies unlikely to be tested in trials
- Post-approval real-world effectiveness, with different populations / protocols than trials
- Prospective observational designs aimed at reducing δ

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Outside medicine: anywhere there's a sharp change in treatment likelihood and you can't lean on no-confounding or parallel trends:

- rolling out a new feature people adopt by choice
- effects of new classes / training programs / curricula opted into
- media or messaging that people consume without randomization

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- SCQE can do work where DID and IV can't (no labeled control group, no clean exclusion).
- The one-cohort form gives you a partial-ID treatment of external control arms.
- Comfort with partial identification takes practice. Reporting ranges instead of points is the point.

Appendix: inference

Consider the regression using the pseudo-outcome $\tilde{Y}_i = Y_i - \delta T_i$:

$$\tilde{Y}_i = \beta_0 + \beta_{\text{SCQE}} D_i + \mu_i,$$

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Alternatives in the package: Anderson–Rubin intervals (no weak-instrument concerns), block-bootstrap over clinics for cluster-robust inference.